Multivariable Linear Regression for Telco Monthly Subscription Costs

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Telco has ten services that contribute to the monthly bill of customers. These services do not have the same financial value for billing customers. In order to determine how the monthly bills are affected by individual services, a multivariable linear regression was done to predict the final monthly charge for a customer depending on which monthly services they subscribe to. Based on this regression model, the base rate for a customer with zero monthly subscriptions is $24.95.

The services and their corresponding predicted upcharges are listed below.

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Service 1: phone_service for $20.05 per month.
Service 2: multiple_lines for $5.02 per month.
Service 3: online_security for $5.05 per month.
Service 4: online_backup for $4.99 per month.
Service 5: device_protection for $5.01 per month.
Service 6: tech_support for $5.02 per month.
Service 7: streaming_tv for $9.98 per month.
Service 8: streaming_movies for $9.95 per month.
Service 9: internet_fo for $24.94 per month.
Service 10: internet_no for $-25.06 per month.
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The most expensive service is fiber-optic internet for ~$24.94 per month. The least expensive service, excluding no internet service, is an online backup service for ~$4.99 per month. For example, if a customer were to add fiber-optic internet and both streaming services, their bill would increase by $(24.94 + 9.98 + 9.95) = $44.87 per month, based on this model.

For the specific bundle subscription (phone service + multiple lines + tech support), the monthly bill would be the flat rate plus these additional charges. The predicted bill would be $55.04/month.

Knowing how each specific streaming service affects the bill is significantly better than just knowing the quantity of additional services. Just two add-ons could be very small or very large, depending on the service. Because of the wide range of regression coefficients, it is more appropriate to fit this with multiple variables. As a concrete comparison, assume two customers have two add-ons. Customer A has phone service and tv streaming, but Customer B has just tech support and device protection. Their bills based on this model are $30.03 additional and $10.03 additional respectively, but they would receive the same bill prediction if the model was based strictly on total number of add-on services.

This regression model has a mean absolute error of $0.80. Based on the testing data, the predictions are, on average, within $0.80 of their actual value. The data boasts an R-Squared value well over 0.99, making the model very accurate.